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Studierichting: Informatica

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$$f(x) = \begin{cases} x & \text{als } x \in [0, \pi] \\ 0 & \text{als } x \in [-\pi, 0] \end{cases}$$

$$a_n = \langle f | \cos(nx) \rangle = \frac{1}{\pi} \int_0^\pi x \cos(nx) dx = \left[\frac{x \sin(nx)}{n\pi} \right]_0^\pi - \frac{1}{n\pi} \int_0^\pi \pi \sin x dx$$

$$= \left[\frac{\cos(nx)}{n^2\pi} \right]_0^\pi = \frac{(-1)^n - 1}{n^2\pi}$$

$$b_n = \langle f | \sin(nx) \rangle = \frac{1}{\pi} \int_0^\pi x \sin(nx) dx = -\frac{1}{n\pi} \int_0^\pi x d(\cos(nx))$$

$$= \left[\frac{x \cos(nx)}{n\pi} \right]_0^\pi + \left[\frac{\sin(nx)}{n^2\pi} \right]_0^\pi = \frac{-(-1)^n}{n} = \frac{(-1)^{n-1}}{n}$$

$$a_0 = \langle f | 1 \rangle = \frac{1}{\pi} \int_0^\pi x dx = \left[\frac{x^2}{2\pi} \right]_0^\pi = \frac{\pi}{2}$$

$$\Rightarrow \hat{f}(x) = \frac{\pi}{4} + \sum_{n=1}^{\infty} \left(\frac{(-1)^n - 1}{n^2\pi} \cos(nx) + \frac{(-1)^{n-1}}{n} \sin(nx) \right)$$

References